Q1. JEE Main 2026 (23 January Shift 1)

Number of solutions of $\sqrt{3}\cos 2\theta + 8\cos\theta + 3\sqrt{3} = 0, \theta \in [-3\pi, 2\pi]$ is:

(1) 0 (2) 5 (3) 3 (4) 4

Q2. JEE Main 2026 (24 January Shift 2)

The number of elements in the set $\{x \in [0, 180^\circ] : \tan(x + 100^\circ) = \tan(x + 50^\circ)\tan x \tan(x - 50^\circ)\}$ is ____.

ANSWERS AND SOLUTIONS

1. (2) **2.** 4

1. **(2)** Using $\cos 2\theta = 2\cos^2\theta - 1$:

$2\sqrt{3}\cos^2\theta + 8\cos\theta + 2\sqrt{3} = 0 \Rightarrow \sqrt{3}\cos^2\theta + 4\cos\theta + \sqrt{3} = 0.$

$\cos\theta = \frac{-4\pm 2}{2\sqrt{3}}$, giving $\cos\theta = -\frac{1}{\sqrt{3}}$ or $\cos\theta = -\sqrt{3}$ (rejected).

Let $\alpha = \cos^{-1}(-1/\sqrt{3}) \approx 2.186$ rad. General solution: $\theta = 2n\pi \pm \alpha$.

In $[-3\pi, 2\pi]$: $\theta = \alpha, -\alpha, 2\pi - \alpha, -2\pi + \alpha, -2\pi - \alpha$.

Total $= 5$ solutions.

2. **(4)** $\frac{\tan(x+100^\circ)}{\tan x} = \tan(x+50^\circ)\tan(x-50^\circ)$

$\frac{\sin(x+100^\circ)\cos x}{\cos(x+100^\circ)\sin x} = \frac{\sin(x+50^\circ)\sin(x-50^\circ)}{\cos(x+50^\circ)\cos(x-50^\circ)}$

Applying Componendo & Dividendo:

$\frac{\sin(2x+100^\circ)}{\sin 100^\circ} = \frac{\cos 100^\circ}{-\cos 2x}$

$2\sin(2x+100^\circ)\cos 2x + \sin 200^\circ = 0$

$\sin(4x+100^\circ) + \sin 100^\circ + \sin 200^\circ = 0$

$\sin(4x+100^\circ) = -2\sin 150^\circ \cos 50^\circ$

$\sin(4x+100^\circ) = -\cos 50^\circ = \sin(-40^\circ)$

$\therefore 4x + 100^\circ = n\pi + (-1)^n(-40^\circ)$

$x = \frac{n\pi + (-1)^{n+1}(40^\circ) - 100^\circ}{4}$

$\therefore x = 30^\circ, 55^\circ, 120^\circ, 145^\circ$ in $(0, \pi)$

$\therefore$ Number of solutions $= 4$